

MineXplore: An Open-Source Reinforcement Learning Exploration Benchmark for GNSS-Denied Underground Environment

Badrikanath Praharaj
VIT Bhopal University
Email: [TBD]

Sreeram M.V.
VIT Bhopal University
Email: [TBD]

[Author 3]
VIT Bhopal University
Email: [TBD]

[Author 4]
VIT Bhopal University
Email: [TBD]

Abstract—We present MineXplore, an open-source, high-fidelity MuJoCo-based simulation testbed for underground robot navigation constructed from the real-world Chilean copper mine survey of Leung *et al.* [1]. Starting from raw survey data, we reconstruct a large-scale 3D tunnel network that preserves key structural properties of the original environment, including irregular tunnel topology, looped connectivity, narrow corridors, and a central rock island. The environment incorporates octagonal wall cross-sections derived from real bore geometry, LiDAR-sourced jagged wall overlays, three spatially distinct terrain friction zones replicating mine water seepage, a global 5-degree incline modelling the tunnel gradient, and periodic warm-yellow spot lighting, all achieving strong geometric consistency with the source map via pixel-to-metric calibration and top-down validation overlays. MineXplore exposes a Gymnasium-compatible navigation interface under local observations, where agents receive egocentric 16-beam LiDAR inputs and control differential-drive motion. To validate that the environment is navigable, we train a baseline single-agent PPO policy and measure coverage, collision rate, and episode length across the full tunnel network. The policy achieves 92.0% coverage of the mapped free space with a 0.14 collision rate over the final 100 evaluation episodes, demonstrating that MineXplore supports reliable policy learning under realistic sensing and topology and providing a reusable open-source benchmark for underground exploration and navigation algorithm development.

Index Terms—simulation environment, underground mining robotics, reinforcement learning, MuJoCo, navigation benchmark, exploration

I. INTRODUCTION

Underground mines are among the hardest environments for autonomous ground robots. GPS is denied, lighting is degraded, dust and water reduce sensor returns, and topology is loop-rich and non-convex [3]. The DARPA Subterranean Challenge surfaced both the difficulty of such settings and the community’s best field systems [2]. Yet much of the simulation tooling used for algorithm development remains centred on Gazebo, even as GPU-accelerated simulators have become the default for modern robot learning pipelines [6], [7].

Two gaps motivate this work. First, no publicly released MuJoCo or MJX environment is grounded in a real production mine; existing subterranean simulation assets are predominantly Gazebo-based [4], [8]. Second, rare published real-mine datasets such as the Leung *et al.* 2017 Chilean underground mine dataset have been used almost exclusively for SLAM

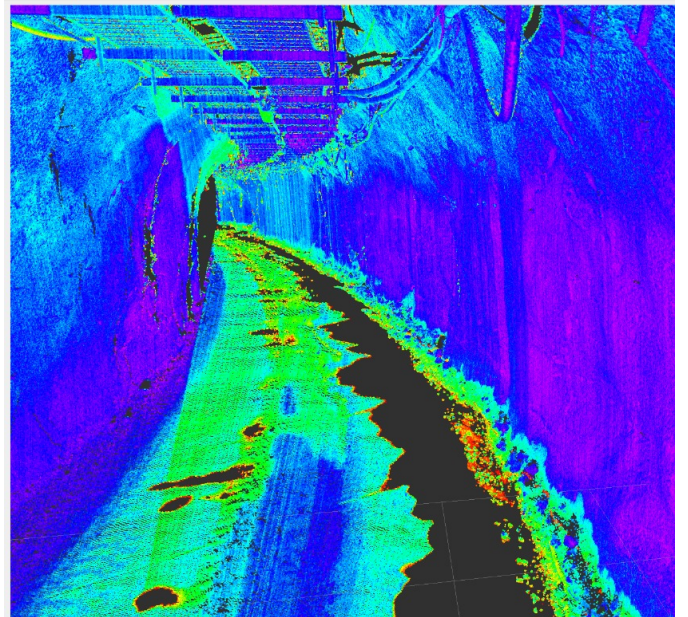


Fig. 1: Point cloud visualisation from the Chilean underground mine survey

evaluation rather than as geometric sources for simulation [1], [3]. MineXplore closes both gaps by compiling the Chilean survey data into a calibrated MuJoCo world directly usable for reinforcement learning under local observations.

Fig. 1 shows a representative point cloud from the Chilean underground mine survey. We present MineXplore, a MuJoCo navigation environment whose geometry is derived from the Leung *et al.* 2017 dataset [1]. The environment exposes a Gymnasium-style reset/step interface and supports training of reinforcement learning policies under local observations in a topologically complex layout that procedural generators such as BARN do not produce [5]. A full description of the contour-to-MuJoCo XML Format (MJCF) compilation and validation pipeline is provided in Section III.

II. RELATED WORK

A. Underground and Subterranean Robotics

The DARPA Subterranean Challenge (2018–2021) drove recent progress in autonomous systems for GPS-denied underground environments, with Team CERBERUS winning the Final Event [2], [4]. Ebadi *et al.* catalogued open problems in extreme underground SLAM across six SubT teams [3]. The Leung *et al.* 2017 Chilean underground mine dataset, our geometric source, remains the only publicly released production-mine sensor log suitable for navigation and SLAM research [1].

B. Simulation Environments for Robot Learning

Gazebo-based assets dominate the subterranean simulation niche: the SubT Virtual Testbed ships replica competition courses [4], and BARN provides 300 procedurally generated navigation worlds used in ICRA benchmarks [5]. For large-scale reinforcement learning, Isaac Gym [6] and MuJoCo Playground [7] have become standard.

C. Prior Use of the Chilean Mine Dataset

Prior work has used the Leung *et al.* dataset exclusively for SLAM benchmarking [3]. To the best of our knowledge, no prior work has compiled its survey geometry into a physics-based simulation environment.

D. Real-Mine-Grounded Simulation

Closest to our work, Gao and Awuah-Offei built a ROS 2 and Gazebo framework for quadruped navigation in 3D maps of real sites including the Edgar Mine teaching facility [8]. MuJoCo, and in particular its GPU-vectorised MJX backend, supports parallel environment stepping at rates orders of magnitude faster than Gazebo, consistent with the throughput expectations of modern RL pipelines [6], [7].

E. Positioning of MineXplore

- **Real production-mine geometry.** Prior assets use procedural generation [5], competition replicas [4], or teaching facilities [8]. MineXplore is the first environment compiled from a real operational mine [1] inside MuJoCo/MJX.
- **Chilean dataset used for simulation.** All prior work uses Leung *et al.* [1] solely for SLAM benchmarking [3]; MineXplore is the first to convert it into a physics-based RL environment.
- **GPU-scalable over Gazebo.** Gazebo-based frameworks [4], [8] are impractical for deep RL training throughput; MuJoCo/MJX provides the parallel stepping rates required [6], [7].
- **Gymnasium interface over real mine topology.** Unlike BARN [5], Isaac Gym [6], or MuJoCo Playground [7], MineXplore exposes a standard reset/step interface over a 104,423 m² loop-rich, real-mine tunnel network.

III. ENVIRONMENT DESIGN

Section III walks through the MineXplore compilation pipeline in order: source inputs, 2D geometry extraction, 3D compilation, surface feature and texture mapping, environment realism features, and validation. Stage labels correspond to Fig. 2.

A. Source Data and Scale Calibration (Stage 1)

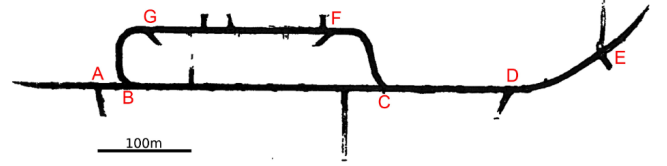


Fig. 3: Published 2D survey floor-plan of the Chilean underground copper mine [1].

MineXplore uses two complementary inputs from the Leung *et al.* 2017 Chilean underground mine dataset [1]. The tunnel geometry is derived from the published 2D survey floor-plan image shown in Fig. 3. The *visual surface features*—wall roughness, cross-sectional profile, and surface colour—are derived from the accompanying 3D LiDAR point cloud logs. We calibrated the pixel-to-metric scale against the 100 m scale bar on the source image, yielding 1.36 px/m. The nominal tunnel height is set to 3.5 m, consistent with the production tunnel specification in the source dataset. The base floor geometry is a flat horizontal plane with no fine-grained elevation mesh; cross-sectional height variation along the tunnel is not modelled at the mesh level. Region-level elevation is introduced separately via a global rotation in Stage 7B, and surface roughness is introduced geometrically in Stage 7A (Section VI).

B. 2D Geometry Extraction (Stages 2–3)

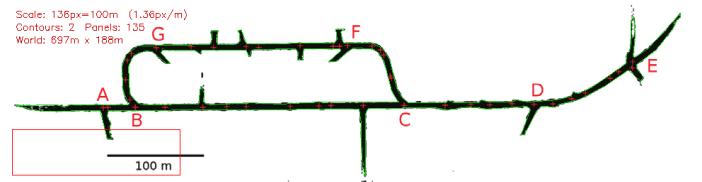


Fig. 4: Detected contours overlaid on the binarised survey map. Outer tunnel boundary in green; interior rock island in red.

We binarised the survey floor-plan and extracted the tunnel footprint using OpenCV contour detection. The largest external contour defines the outer tunnel boundary; interior contours define rock islands and pillars. Fig. 4 shows the detected contours overlaid on the binarised map. Each contour was converted to a Shapely polygon with holes and simplified using Douglas–Peucker with a tolerance tuned to preserve

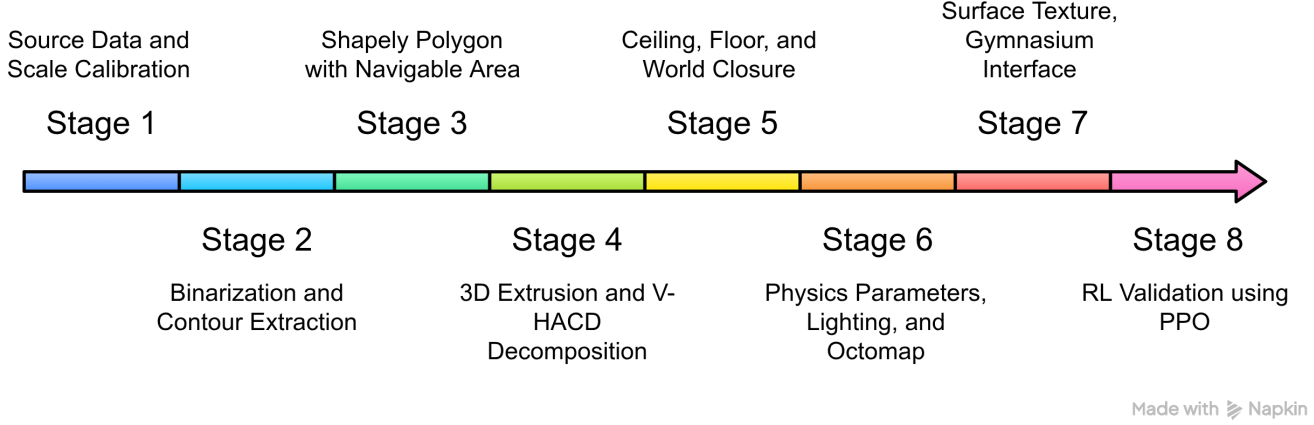


Fig. 2: MineXplore compilation pipeline: from real Chilean mine survey data through eight stages

corners under 1 m. The natural irregularity retained by this tolerance is the primary source of jagged wall geometry in the compiled model. The compiled footprint contains one outer boundary and one dominant interior rock island of 14,577 m², yielding a navigable area of 104,423 m² within a world extent of approximately 697 m × 188 m.

C. 3D Compilation and Physics Setup (Stages 3–6)

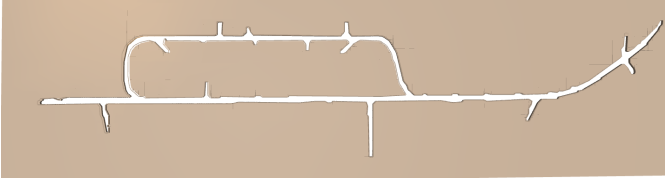


Fig. 5: Compiled MineXplore environment in the MuJoCo interactive viewer.

We triangulated the 2D polygon set and extruded each face to 3.5 m (Stage 4). MuJoCo’s collision engine requires convex geoms; we decomposed the extruded mesh using V-HACD with a resolution of 1,000,000, a concavity threshold of 0.5, and a maximum of 32 convex hulls per rock tile. Applied across 70 rock geometry tiles, this produces 1,186 convex collision geoms in the final MJCF model. The compiled world is loaded and is centred on the main tunnel network. Ceiling and floor planes (Stage 5) close the tunnel volume. Physics parameters (Stage 6) set baseline wall and floor friction and include an occupancy octomap for collision-free spawn sampling. Compilation and scene load complete in under ten seconds on an AMD Ryzen 5 5600H under Ubuntu Linux. Fig. 5 shows the compiled MuJoCo world.

D. Surface Wall Geometry and Visual Detail (Stage 7A)

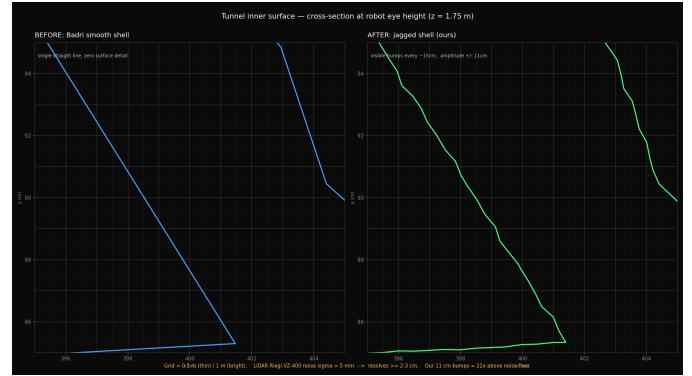


Fig. 6: Cross-sectional wall profile comparison: smooth flat-extruded geometry (left) versus the two-layer surface model

Real copper mine tunnels are mechanically bored and blasted, producing wall cross-sections that are roughly polygonal rather than rectangular, with highly irregular surfaces at the centimetre scale. MineXplore replicates both properties using a two-layer visual wall system applied on top of the V-HACD collision geometry.

The *base layer* is, an 8-sided polygonal mesh that approximates the characteristic bore cross-section of the production tunnel. The *overlay layer* is, a jagged mesh derived from the Leung *et al.* 3D LiDAR point cloud that captures the centimetre-scale rock surface irregularity of the real tunnel walls. Both mesh geoms are configured as purely visual and do not participate in collision detection; physics contact is handled exclusively by the V-HACD convex hull decomposition from Stages 3–6. This two-layer design separates physical accuracy from visual realism and avoids the computational cost of simulating fine-surface contact. A uniform rock material is applied across all 1,186 rock meshes and both wall

layers. Fig. 6 illustrates the difference between the original flat-extruded geometry and the two-layer surface model.

E. Environment Realism Features (Stage 7B)

MineXplore incorporates three additional features that replicate the physical conditions of the Chilean copper mine.

Terrain incline. The entire MJCF worldbody is rotated 5° (0.0872665 rad) around the Y-axis, tilting walls, floor, rocks, and lights uniformly as a rigid body. This produces an effective elevation gain of 5.2 m across the tunnel’s entrance region, consistent with the terrain gradient in the source dataset, without modifying the V-HACD collision decomposition.

Terrain friction zones. Three spatially fixed floor box geoms partition the tunnel along its east-west axis to replicate mine water seepage: a dry western zone (friction 1.00), a muddy central zone (friction 0.55), and a waterlogged eastern zone (friction 0.25), each covering $119 \text{ m} \times 102 \text{ m}$. The contact friction compounds with the actuation-level scaling in Section IV to jointly model traction degradation across ground conditions.

Periodic lighting. Thirty-three MuJoCo spot lights are placed along the tunnel centreline at 20 m intervals, 3.0 m above the floor, directed slightly downward with a 45° cone cutoff.

F. Geometry Validation

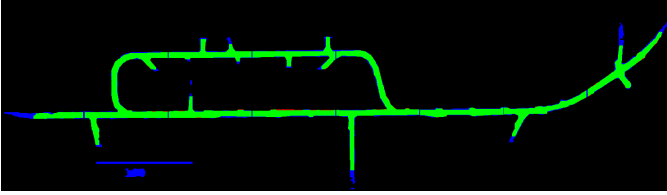


Fig. 7: Pixel-wise geometric fidelity overlay. Green pixels indicate agreement and blue pixels indicate deviation

We validated the compiled model on two axes. For *navigability*, we cast rays through the free volume and confirmed that every pixel marked navigable in the source floor-plan maps to an unobstructed column in the MJCF model. For *geometric consistency*, we rendered an orthographic top-down projection of the compiled MJCF model, binarised it, and compared it pixel-wise against the original survey floor-plan. Fig. 7 shows the resulting overlay: green pixels mark agreement; blue pixels mark deviation, concentrated at junctions and corridor edges where the Douglas–Peucker simplification rounds tight corners. The strict Intersection over Union (IoU) after noise cleanup is **0.9538**, confirming that the compiled model reproduces 95.3% of the source survey footprint at pixel level. This overlay is produced directly from the compiled MJCF, not from any intermediate representation.

G. Surface-Texture Validation

3-D LiDAR point cloud by fitting SVD-based principal planes over a voxel-scale sweep $w \in \{0.10, \dots, 2.00\} \text{ m}$, computing ISO 4287-style residuals (R_a , R_q , R_z , p_{95}), and

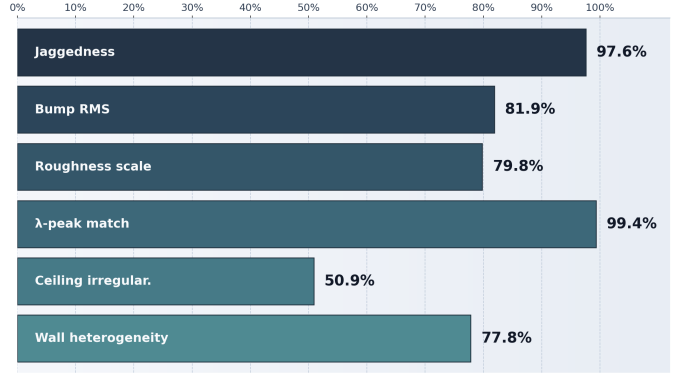


Fig. 8: Per-axis surface-texture similarity comparison

extracting a 1-D along-wall wavelength spectrum via arc-length-resampled FFT across 5 cm horizontal slabs. Six structural dimensions were scored on a per-axis similarity metric (Fig. 8): wall jaggedness (97.6%), bump R_q (81.9%), roughness scale integral (79.8%), dominant wavelength λ -peak (99.4%), ceiling irregularity (50.9%), and wall heterogeneity (77.8%), yielding an overall geometric-mean score of **79.4%**. The simulated surface under-shoots real roughness magnitude by $\approx 2.5\times$ at the half-metre scale — an intentional amplitude clip to prevent narrow passages from pinching shut — while faithfully replicating the dominant 5–20 cm spatial-frequency band of the real mine walls.

IV. BENCHMARK INTERFACE

A. Observation Space

Each agent receives a 19-dimensional observation vector at every control step. The first 16 elements are LiDAR range readings from a simulated 2D scanner with 16 uniformly spaced beams covering the full 360° field of view $([0, 2\pi])$. Raw ranges are clipped to $[0.12 \text{ m}, 10.0 \text{ m}]$, then normalised to $[0, 1]$. Gaussian noise with $\sigma = 0.01 \text{ m}$ is applied to each beam before normalisation. The remaining 3 elements are: the previous normalised linear command, a fixed scalar of 0.0, and the previous normalised angular command, providing the agent with one step of action history.

B. Action Space

The action space is a continuous 2D box $\mathcal{A} = [-1, 1]^2$. The policy output (a_0, a_1) is mapped to differential-drive commands as:

$$v = a_0 \cdot 1.5 \text{ m/s}, \quad \omega = a_1 \cdot 1.5 \text{ rad/s} \quad (1)$$

Commands are then scaled and perturbed by terrain type before execution: dry terrain applies scale 1.0 with zero noise; muddy terrain applies scale 0.7 with additive noise $\sigma = 0.04 \text{ m/s}$; waterlogged terrain applies scale 0.4 with noise $\sigma = 0.10 \text{ m/s}$. This actuation-level scaling compounds with the MuJoCo contact friction values described in Stage 7B to model the full traction degradation across terrain types.

C. Episode Structure

MineXplore is a pure exploration environment. There is no goal position: `terminated` is always `False` and episodes end only by truncation. The maximum episode horizon is 6,000 control steps at a timestep of 0.1 s, giving a nominal maximum duration of 600 s. An episode is also truncated early if the agent’s speed remains below 0.01 m/s for 100 consecutive steps, preventing degenerate stationary policies. At the start of each episode, the agent is spawned at a fixed anchor near the mine entrance; if that cell fails clearance checks, the agent falls back to a random valid point in the entrance region with uniform yaw sampling over $[-\pi, \pi)$.

D. Reward Structure

The per-step reward r is:

$$r = \Delta c \cdot g - \mathbb{I}[\Delta c = 0] \cdot 0.002 + M - \mathbb{I}[\text{collision}] \cdot 15.0 \quad (2)$$

where Δc is the number of newly discovered free-space cells at the current step, and $g = 10.0/C_{\text{total}}$ is a per-cell gain normalised by the total number of tunnel cells C_{total} . A small step penalty of -0.002 is applied on steps where no new cells are discovered to discourage stationary behaviour. One-time milestone bonuses M are awarded at 25% coverage (+15), 50% (+30), and 75% (+45). A collision penalty of -15.0 is applied on every step in which the agent contacts a non-floor geometry. We use reinforcement learning solely to validate that MineXplore supports stable policy learning and meaningful exploration under local observations; RL is not the primary contribution of this paper.

E. Reproducibility

In the experiments reported here we use a single-agent configuration. When multiple agents are active, each agent involved in an inter-agent collision receives the full -15.0 collision penalty independently; uninvolved agents are unaffected. All experiments use MuJoCo 3.2.6 with fixed random seeds and the compiled MJCF model under version control. Code and environment pipeline will be made open-source upon publication.

V. RESULTS

A. Environment Validation

The pixel-wise overlay of Fig. 7 confirms geometric fidelity at a strict IoU of **0.9538** between the compiled MJCF footprint and the original survey map; residual deviation (blue pixels) is localised to junctions and corridor edges, consistent with the Douglas–Peucker simplification tolerance. Ray-cast navigability passes for 100% of sampled free-space points, confirming that no navigable pixel in the source floor-plan is occluded by collision geometry in the compiled model.

B. Reinforcement Learning Baseline

We use a single-agent PPO policy to validate that MineXplore supports stable exploration learning under local observations. RL is not a contribution of this paper; the baseline exists solely to confirm that the environment is navigable and that reward signals are well-formed. Training ran for 254 episodes on a single agent with the reward structure defined in Section IV. All metrics below are reported as means over the final 100 episodes.

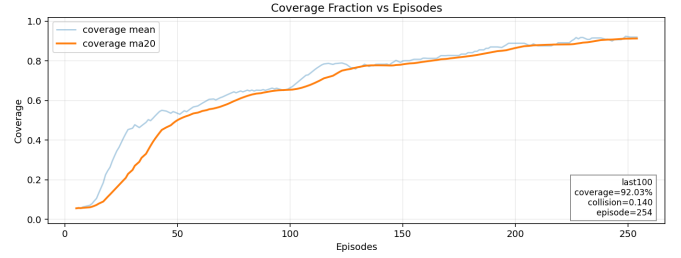


Fig. 9: Coverage fraction over training.

Fig. 9 shows the fraction of total navigable tunnel area discovered per episode over training. Coverage rises from approximately 5% at episode 1 to **92.03%** in the final 100 episodes. The smooth, monotonic increase in the moving average (ma20) with no plateau before episode 150 confirms that the environment presents a non-trivial exploration challenge that rewards continued learning rather than saturating early.

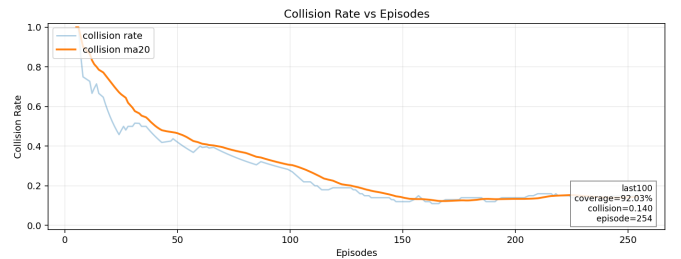


Fig. 10: Collision rate over training.

Fig. 10 shows the per-episode collision rate over training. The rate begins near 1.0 in early episodes, reflecting an untrained policy that contacts walls frequently in the narrow, irregular corridors. It falls steadily to **0.140** in the final 100 episodes, demonstrating that the agent learns to navigate within the V-HACD collision geometry without systematic failures.



Fig. 11: Cumulative episode reward over training.

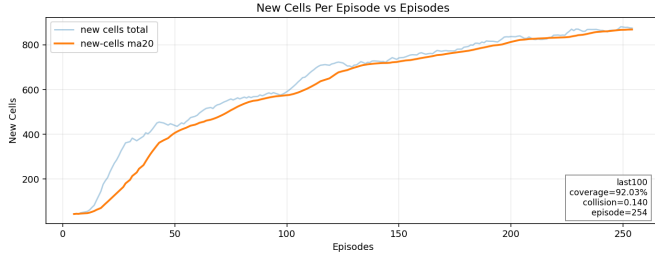


Fig. 12: New cells discovered per episode over training.

Figs. 11 and 12 show cumulative episode reward and new cells discovered per episode respectively. Both metrics rise monotonically from negative reward (≈ -300) and ≈ 75 new cells per episode at the start, to $\approx 2,050$ cumulative reward and ≈ 860 new cells by episode 254. The alignment between these two curves directly reflects the reward structure: new-cell discovery is the primary reward signal, so reward growth is structurally coupled to exploration breadth. This coupling confirms that the reward function is well-calibrated and that no degenerate reward exploitation occurs.

VI. DISCUSSION

A. What MineXplore Is and Is Not

MineXplore is a single-mine MuJoCo/MJX navigation environment grounded in a real production tunnel, targeted at researchers who need a non-procedural, topology-rich underground test case for navigation and exploration algorithm development. It is not a physics-calibrated digital twin: wall friction, surface roughness, and contact stiffness are approximations derived from the source data, not direct measurements.

B. Limitations

the collision model is a V-HACD convex decomposition with resolution 1,000,000 and concavity threshold 0.5; this may introduce localised hull artefacts in the narrowest tunnel passages. The elevation model is a single global rotation approximating the tunnel gradient; fine-grained topographic variation and cross-sectional height changes along the full tunnel length are not modelled and are deferred to future work using the Leung *et al.* 3D LiDAR elevation logs.

C. When to Use MineXplore

MineXplore is the right tool when you need a large-scale, real-mine-derived obstacle topology inside a MuJoCo or MJX training pipeline, a Gymnasium-compatible interface for single- or multi-agent navigation under local observations, or matched source data for future sim-to-real work grounded in the Leung *et al.* Chilean mine dataset [1].

VII. CONCLUSION

We presented MineXplore, an open-source MuJoCo navigation environment derived from the Leung *et al.* 2017 Chilean underground copper mine survey. We described the eight-stage contour-to-MJCF compilation pipeline (Fig. 2), validated the compiled geometry against the source survey via a top-down overlay (Fig. 7), and confirmed navigability using a single-agent PPO baseline that achieves 92.03% tunnel coverage with a 0.140 collision rate over the final 100 training episodes. MineXplore closes a specific gap in the robot-learning tooling landscape: a real-mine-grounded, Gymnasium-compatible navigation environment in the MuJoCo ecosystem, augmented with LiDAR-sourced wall geometry, terrain heterogeneity, a global incline, and periodic spot lighting. Future work includes full elevation-aware geometry from the Leung *et al.* 3D LiDAR logs, LiDAR-replay validation against the original dataset sweeps, procedural cross-section perturbation for environment diversity, and broader multi-agent and cross-algorithm baselines.

ACKNOWLEDGMENT

The authors thank Leung *et al.* for releasing the Chilean underground mine dataset.

REFERENCES

- [1] K. Y. K. Leung, D. Lühr, H. Houshiar, F. Inostroza, D. Borrmann, M. Adams, A. Nüchter, and J. Ruiz-del-Solar, "Chilean underground mine dataset," *The International Journal of Robotics Research*, vol. 36, no. 1, pp. 16–23, 2017. DOI: 10.1177/0278364916679497.
- [2] M. Tranzatto, T. Miki, M. Dharmadhikari, L. Bernreiter, M. Kulkarni, F. Mascari, O. Andersson, S. Khattak, M. Hutter, R. Siegwart, and K. Alexis, "CERBERUS in the DARPA Subterranean Challenge," *Science Robotics*, vol. 7, no. 66, p. eabp9742, 2022.
- [3] K. Ebadi, L. Bernreiter, H. Biggie, G. Catt, Y. Chang, *et al.*, "Present and future of SLAM in extreme environments: The DARPA SubT Challenge," *IEEE Transactions on Robotics*, vol. 40, pp. 936–959, 2024.
- [4] Open Robotics, "DARPA SubT Virtual Testbed," 2021. [Online]. Available: <https://github.com/osrf/subt>
- [5] D. Perille, A. Truong, X. Xiao, and P. Stone, "Benchmarking metric ground navigation," in *Proc. IEEE Int. Symp. Safety, Security, and Rescue Robotics (SSRR)*, 2020.
- [6] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac Gym: High performance GPU based physics simulation for robot learning," in *NeurIPS Datasets and Benchmarks Track*, 2021.
- [7] K. Zakka, B. Tabanpour, Q. Liao, M. Haiderbhai, S. Holt, J. Y. Luo, A. Allshire, E. Frey, K. Sreenath, L. A. Kahrs, C. Sferrazza, Y. Tassa, and P. Abbeel, "MuJoCo Playground," *arXiv preprint arXiv:2502.08844*, 2025.
- [8] Y. Gao and K. Awuah-Offei, "Navigation in underground mine environments: A simulation framework for quadruped robots," in *Proc. IEEE Int. Conf. Automation Science and Engineering (CASE)*, 2025, pp. 1464–1469.